Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/28/2006
PI Name:	Jacobs, Donald T. Ph.D.		
Project Title:	Turbidity and Universality Around a Liquid-Liquid Critical Point		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature condensed matter physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	44691	**Congressional District:**	16
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	96-HEDS-03
Start Date:	02/01/1998	**End Date:**	01/31/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	24	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	None		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock the mysteries of the universe obscured by Earth's gravity		
Task Description:	In order to avoid the difficulties of gravity and multiple-scattering associated with other experiments, our turbidity experiment investigates a liquid-liquid mixture (methanol-cyclohexane) whose densities are closely matched. Such a binary liquid mixture avoids the gravity effects of concentration gradients and beam bending which are so troublesome in Earth-bound experiments on a liquid/gas system. The turbidity will be measured to avoid problems with multiple scattering, as occurs in angular intensity measurements. The turbidity provides a measure of the critical opalescence of the system, which is caused by light scattering from droplets. By analyzing the transmitted light intensity and using current theoretical predictions, the value of the critical exponent eta can be determined. Measuring the turbidity determines three interrelated critical exponents and can be used to test the theoretical scaling relation gamma = (2 - eta)*nu. In addition, two measurable amplitudes can be combined with the heat capacity amplitude to test two-scale factor universality.		

Task Description:	This ground-based experiment investigates the light scattered by a simple, density matched, liquid-liquid mixture which is an excellent model system. Our investigation will test existing theory for the value of the critical exponent eta by measuring the experimental value from the total light being scattered (turbidity) in the binary fluid mixture methanol and cyclohexane. The resulting data will be used to confirm and quantify a small exponent "eta" in the expression {(1+q*xi)^(2?eta)} that describes the correlation length dependent distribution of scattered light from near critical fluids. Also, "two-scale universality" can be tested from the data set when combined with measurements of the heat capacity. The universal behavior near a critical point means that the same phenomena should be observed in a wide range of systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)